Commitment of Traders reports in oil futures: a complete analytical guide

The CFTC's Commitment of Traders report remains the single most important public window into who holds what in petroleum futures—but extracting reliable trading signals from it demands navigating classification quirks, structural breaks, and a 3-day data lag that has tripped up countless backtests. The Disaggregated COT report, introduced in September 2009, divides petroleum futures positioning into four trader categories and covers contracts from NYMEX WTI (code 067651, ~2 million contracts open interest) down to niche crack spreads with barely 2,000 lots. Academic research overwhelmingly finds that COT positioning responds to prices rather than predicting them—managed money positions are roughly **40% explainable by simple trend-following signals** (Boos and Grob, 2023)— ScienceDirect yet practitioners continue to find value in positioning extremes as contrarian indicators when combined with fundamental catalysts. This report synthesizes the mechanics, contracts, academic literature, signal construction methods, and oil-market-specific dynamics needed to use COT data rigorously.

Four report types serve different analytical purposes

The CFTC publishes four distinct COT report formats, each with different trader classifications, coverage, and history. For petroleum analysis, the **Disaggregated report** is the primary tool.

The **Legacy report** is the oldest format, with weekly data available back to January 1986. Dirnea Commodity Futures Tradin... It classifies traders into just two categories—commercial and non-commercial— CFTC based on whether they file CFTC Form 40 attesting to bona fide hedging activity under Regulation 1.3(z). Its critical limitation: swap dealers hedging OTC positions in futures were classified as "commercial," lumping Goldman Sachs's commodity desk with an independent Oklahoma crude producer. CFTC This contamination motivated the creation of newer report formats.

The **Supplemental (CIT) report**, introduced January 2007, added an "Index Traders" category but covers **only 13 agricultural commodities**— Dirnea Commodity Futures Tradin... it is irrelevant for petroleum analysis. The **Disaggregated report**, Commodity Futures Tradin... first published September 4, 2009, Commodity Futures Tradin... with historical data backcasted to June 2006, Commodity Futures Tradin... is the workhorse for physical commodity positioning analysis. Barchart It splits traders into four categories: **Producer/Merchant/Processor/User** (PMPU), **Swap Dealers**, **Managed Money** Commodity Futures Tradin... (registered CTAs, CPOs, and CFTC-identified funds), and **Other Reportables**. Commodity Futures Tradin... Classification is based on predominant business activity as self-reported on Form 40 and reviewed by CFTC staff. Commodity Futures Tradin... Commodity Futures Tradin... A single entity can be classified differently across different commodities— Commodity Futures Tradin... Barchart a bank might be a swap dealer in crude oil but managed money in equities. Commodity Futures Tradin... The **Traders in Financial**

Futures report covers only financial contracts (currencies, treasuries, equity indices) and is not applicable to petroleum. TradingView +2

The reporting schedule follows a fixed cadence: positions are **snapshotted each Tuesday at market close**, firms report to CFTC on Wednesday morning, and the data is published **Friday at 3:30 PM Eastern Time.** Barchart +4 This creates a minimum 3-business-day lag. Cotdata Federal holidays delay release by one to two days Commodity Futures Tradin... roughly five times per year. More disruptive interruptions have occurred: the ION cybersecurity incident in February 2023 delayed reports for several weeks, Commodity Futures Tradin... and a federal appropriations lapse in late 2025 interrupted publication until November 12, 2025, Cot-reports with the backlog not cleared until January 23, 2026. Commodity Futures Tradin...

Non-reportable positions—those below CFTC reporting thresholds—are calculated as a residual: total open interest minus all reportable positions. Barchart Commodity Futures Tradin... The thresholds for petroleum are **350 contracts for crude oil** (350,000 barrels), **250 for heating oil/ULSD**, and **150 for RBOB gasoline**. Once any single contract month triggers the threshold, all of a trader's positions in that commodity become reportable. Springer +2 Reportable positions typically cover **70–90% of total open interest.** Barchart Wikipedia

Petroleum contracts span three liquidity tiers across NYMEX and ICE

The CFTC's Disaggregated Petroleum and Products report covers over 25 contracts. The nine most liquid—all with open interest exceeding 200,000 contracts—form the core for systematic analysis:

Contract	CFTC Code	Exchange	Approx. OI (Futures)	Traders
WTI-Physical (CL)	067651	NYMEX	2,002,000	307
ICE WTI	067411	ICE Europe	846,000	122
WTI Houston Argus (HCL)	0676A5	NYMEX	464,000	—
RBOB Gasoline (RB)	111659	NYMEX	355,000	236
Brent Last Day (BZ)	06765T	NYMEX	316,000	68
WTI Midland Argus	067A71	NYMEX	303,000	—
NY Harbor ULSD (HO)	022651	NYMEX	256,000	157
WTI Financial (CSW)	06765A	NYMEX	226,000	—
WTI-Brent Calendar	06765L	NYMEX	207,000	—

ICE Brent Crude and ICE Gasoil are not in the CFTC COT report. They are traded on ICE Futures Europe (UK-regulated) and covered only by ICE's own COT report, published Fridays at 18:30 London time (ICE) using identical Disaggregated categories. (ice) This means comprehensive oil market positioning analysis requires merging two data sources: CFTC data for WTI and refined products, and ICE Europe data for Brent and gasoil.

The participant mix differs markedly between exchanges. NYMEX WTI shows diversified participation: PMPUs hold roughly 29% of longs and 19% of shorts, swap dealers account for about 6.5% of longs but **30.4% of shorts**, and managed money holds about 9% of longs. (Commodity Futures Tradin...) (cftc) ICE WTI is far more dominated by producers—PMPUs hold **59.8% of longs and 46.3% of shorts**— (Commodity Futures Tradin...) (cftc)reflecting its heavier use by physical commodity participants operating through London. For headline speculative positioning analysis, practitioners should aggregate managed money net positions across NYMEX and ICE WTI (since ICE WTI is ~42% of NYMEX WTI's open interest, omitting it materially understates total speculative activity). For market microstructure analysis, keeping them separate is warranted given the different trader compositions.

Academic evidence finds positioning follows prices, with important nuances

The research literature on COT predictive power spans three decades and converges on a central finding: **positions primarily respond to price changes rather than causing them**, but hedging pressure dynamics create measurable risk premia.

De Roon, Nijman, and Veld (2000, *Journal of Finance*) established the hedging pressure hypothesis, (Tilburg University) showing that futures risk premia depend on both own-market and cross-market hedging pressures (ScienceDirect) across 20 futures markets. (Wiley Online Library) Kang, Rouwenhorst, and Tang (2020, *Journal of Finance*) refined this into a two-premium framework: (Semantic Scholar) a **short-term liquidity premium** where momentum-driven speculators provide liquidity and trade impatiently, and a **long-term hedging pressure premium** (Wiley Online Library) where net producer selling pressure creates predictable upward drift. (Carloalberto) Money managers are predominantly positive-feedback (momentum) traders; producers are contrarian negative-feedback traders. (Carloalberto) (UC)

The most direct test of predictive power comes from Sanders, Irwin, and Merrin (2009, *Journal of Agricultural and Resource Economics*), who applied bivariate Granger causality tests to 10 agricultural futures and found **"very little evidence"** that COT positions forecast returns. (Springer) Sanders, Boris, and Manfredo (2004, *Energy Economics*) reached the same conclusion specifically for crude oil, gasoline, heating oil, and natural gas: positive returns lead to increased non-commercial net longs, not vice versa. (Springer) Büyükşahin and Harris (2011, *The Energy Journal*), using non-public daily CFTC data for crude oil specifically, found no Granger-causal link from speculative positions to price changes. (ScienceDirect +2)

The critical analytical breakthrough came from Boos and Grob (2023, *Journal of Financial Markets*), who decomposed speculative positioning into trend-following and residual components using ridge regression. They found that **trend-following explains over 40% of the variance in speculator position changes** across 23 commodities, with the trend signal remarkably stable across markets and time. ScienceDirect This means the majority of what COT data shows about managed money positioning is simply a reflection of recent price action—not independent information. The **residual component** (the roughly 60% not explained by trend-following) potentially contains genuine conviction-based positioning, but isolating and trading on this residual remains an active area of research. Uhl (2025, *Review of Financial Economics*) extended this work to show that when speculative crowding is high—many CTAs aligned with generic trend signals—subsequent trend-following performance deteriorates. Wiley Online Library

On the question of whether speculators destabilize markets, Brunetti, Büyükşahin, and Harris (2016, *JFQA*) found that hedge fund position changes are **negatively** correlated with volatility in crude oil, natural gas, and corn—hedge funds provide liquidity and facilitate price discovery. Bank of Canada The ECB's 2024 Economic Bulletin quantified the amplification effect: the elasticity of crude oil spot prices to supply shocks increases from **1.6 when speculative positions are low to 1.8 when they are large**, European Central Bank but the confidence intervals overlap—the difference is not statistically significant. European Central Bank

Dewally, Ederington, and Fernando (2013, *Review of Financial Studies*) provide the strongest energy-specific evidence, using proprietary CFTC data on crude oil, gasoline, and heating oil traders. They found that **mean hedger profits are negative while speculator profits are positive** (supporting the risk premium hypothesis), profits on long positions vary inversely with inventories and directly with volatility (supporting the theory of storage), and momentum in commodity futures may be driven largely by hedging pressure. SSRN SSRN

Constructing signals requires normalization, confirmation, and discipline about lag

The most widely used COT signal is the **COT Index**, a min-max normalization pioneered by Larry Williams in the 1970s:

COT Index = (Current Net Position − Min over N weeks) / (Max over N weeks − Min over N weeks) × 100

The standard lookback is **156 weeks (3 years)** for swing trading signals, Scribd with 52 weeks as an intermediate alternative and 26 weeks for Williams's OI-adjusted "WillCo" variant. MQL5 Signal thresholds range from 75/25 (Williams) Williamspercentr to 90/10 (higher-conviction practitioners like Martin Goersch). Wang's (2001, *Journal of Futures Markets*) academic sentiment index uses an identical formula and found that extreme hedger bullishness predicts price reversals while extreme speculator bullishness predicts continuations— ResearchGate a crucial asymmetry for signal design.

The **z-score approach** (current net position minus rolling mean, divided by rolling standard deviation) is more statistically rigorous and less sensitive to single outlier observations. A |z| > 2.0 identifies approximately 95th/5th percentile positioning. TradingView **Percentile rankings** over rolling windows offer similar robustness. Normalizing net positioning by total open interest—yielding a net-position-to-OI ratio—adjusts for secular market growth Cotunchained and is essential for comparing signals across different time periods when crude oil open interest has expanded dramatically.

Jason Shapiro, profiled in Schwager's *Unknown Market Wizards* Elearnmarkets with a track record of **34% annualized returns at 20% target volatility**, Analyzing Alpha uses COT extremes combined with "news failure" confirmation: he identifies crowded speculative positioning, waits for news that should reinforce the trend, and enters when price moves opposite to expectations. Elearnmarkets His philosophy—"positioning is the ultimate discounting mechanism"—frames COT data not as a predictive tool but as a map of where pain trades will occur when catalysts arrive. Crowdedmarketsreport

The most important validation consideration is **point-in-time fidelity**: signals must be acted on no earlier than Friday's close or Monday's open, since Tuesday's snapshot is not available until Friday at 3:30 PM ET. PyPI Cotdata Any backtest using Tuesday data before Friday introduces look-ahead bias. Bookdown Rolling normalization windows must use only past data at each point—computing z-scores or percentile ranks over the full sample is a common but invalidating error. Quantreo The 2009 structural break (Disaggregated report introduction) means pre-2009 "commercial" positions in Legacy reports are contaminated by swap dealer activity; Amazon UK Commodity Futures Tradin... research spanning this boundary must either use only Legacy classifications throughout or acknowledge the break explicitly.

Dreesmann, Herberger, and Charifzadeh (2023, SSRN #4407250) applied Monte Carlo simulation to COT reversal strategies and found that while individual markets showed statistically significant results, **portfolio-level implementation underperformed**— SSRNa sobering reminder of selection bias in single-market COT research. The weekly frequency (52 observations per year) severely limits statistical power, making overfitting detection difficult and requiring minimum 10-year histories for meaningful validation.

Multi-factor combinations improve robustness. Fuertes, Miffre, and Rallis (2010) demonstrated that sorting commodities jointly by momentum and term structure (carry) signals produced **21% annualized alpha**, roughly double what either signal achieved alone. ScienceDirect Adding COT positioning as a third factor—particularly hedging pressure measures derived from PMPU net positions—provides a distinct risk factor with low correlation to momentum and carry (Basu and Miffre, 2013). The practical implementation combines COT extremes as a directional filter with daily-frequency momentum or RSI for entry timing.

Oil market dynamics create unique positioning patterns

OPEC+ supply management is the most powerful external driver of oil positioning shifts. The OPEC Monthly Oil Market Report tracks these dynamics closely. In June 2024, managed money net longs in combined ICE Brent and NYMEX WTI rose **28%**, with speculators buying the equivalent of 78 million barrels as Middle East tensions escalated. OPEC In August 2025, the dynamic reversed sharply: managed money moved net short in WTI, selling the equivalent of **154 million barrels in a single week** following OPEC+'s surprise decision to increase production by three times more than initially planned. European Central Bank OPEC These positioning swings—often 20-30% of net positions within weeks—dwarf movements in most other commodity markets.

The **reflexive feedback loop** between positioning and price operates asymmetrically. Position building is gradual (momentum-driven accumulation over weeks), while unwinds can be violent and self-reinforcing. EBC Financial Group Investing.com The October 2025 Gaza ceasefire triggered an immediate positioning unwind as "the removal of fear-based pricing allowed fundamental market factors to reassert their influence." Investing.com Daman Markets Options-driven gamma effects amplify this asymmetry: when large option positions sit near the money, hedging flows create feedback loops where "price moves trigger hedging, hedging moves price, and volatility either compresses or accelerates." OilPrice.com The Oxford Institute for Energy Studies (December 2025) found **no correlation between money manager position levels and realized crude volatility**—in fact, volatility was highest when fund positioning was at its lowest, Oxford Institute for Energy… consistent with the view that extreme unwinds from crowded positions, not the positions themselves, generate volatility.

Seasonal patterns in petroleum positioning are well-documented and directly tied to the refinery cycle. Moore Research Center's 15-year analysis shows crude oil typically bottoms in December and begins a spring advance, with May crude closing higher on February 24 versus February 6 in **87% of years** (average gain: $2.26/barrel). Atkinsongrain Farmerswin Commercial traders display predictable seasonal buying in Q1 as refiners accumulate crude ahead of summer driving season Atkinsongrain and the EPA's May 1 deadline for summer-grade gasoline at terminals. Inkl In early 2026, COT data showed commercials at their **least bearish reading in 12 months**, confirming this seasonal pattern. Atkinsongrain Farmerswin

The **shale revolution** fundamentally altered the producer/merchant category in COT data. U.S. petroleum production more than doubled from 9.1 million bpd in 2009 to approximately 23 million bpd by 2025. OilPrice.com Producer hedging peaked in 2018 both in absolute terms and as a percentage of production, then declined through 2023 as industry consolidation, ample revenues, steep backwardation, and a shift toward capital returns over growth reduced hedging appetite. CME Group By Q2 2019, only **8% of 2020 production** from large-cap E&P companies was hedged. Dallas Fed However, CME Group data shows hedging activity began recovering in H2 2024, with WTI options volume up **40% year-over-year** in H1 2025 and WTI CMA futures

open interest hitting a post-pandemic high—though the hedging toolkit shifted from complex three-way collars to simpler outright put buying. CME Group

Commodity index traders (CITs) peaked in influence around 2008, when total commodity index investment reached an estimated **$161–174 billion**, Commodity Futures Tradin… CFTC with crude oil commanding over 40% of the S&P GSCI by dollar weight. ScienceDirect The Goldman Roll—the 5th-to-9th-business-day monthly roll of S&P GSCI positions—initially imposed significant costs: CFTC index investors "forwent on average 3.6% annual return" due to roll impact (Yu, CFTC study). But Irwin, Sanders, and Yan (2022) found that roll costs declined by **over 80%** from 1991–2011 to 2012–2019, as markets adapted through a "sunshine trading effect" Scotthirwin (the roll schedule is publicly known). The Masters Hypothesis—that index investors caused the 2007-2008 commodity price surge—has been largely rejected by academic research. Irwin and Sanders (2012, *Energy Economics*) found "very little evidence" linking index positions to returns or volatility across 19 commodity markets. ScienceDirect Cabi Digital Library The CFTC discontinued its Index Investment Data reports in November 2015. ScienceDirect

Conclusion

COT data is best understood not as a crystal ball but as a **positioning map**—it reveals where market participants are concentrated and therefore where forced liquidation or short-covering will occur when catalysts arrive. The academic consensus is clear: positions follow prices, managed money is largely trend-following, and Granger causality runs from returns to positioning rather than the reverse. Springer +2 Yet the framework retains practical value precisely because it identifies the conditions for pain trades. The strongest evidence supports using COT extremes as a **necessary but not sufficient** condition for contrarian trades, combined with fundamental catalysts (OPEC+ decisions, inventory surprises, seasonal transitions) and momentum or term structure signals for timing.

Three implementation principles emerge from the evidence. First, use the Disaggregated report's managed money category rather than the Legacy report's non-commercial classification — Barchart the swap dealer contamination issue makes pre-2009 Legacy commercial data unreliable. Amazon +2 Second, normalize positioning using z-scores or percentile ranks with rolling (not full-sample) windows, and always respect the Tuesday-to-Friday publication lag in backtesting. Third, for petroleum specifically, aggregate NYMEX and ICE WTI managed money positions for a complete picture of speculative activity, and supplement with ICE Europe's own COT report for Brent positioning. The weekly frequency and limited statistical power of COT data mean it works best as one input among several in a multi-factor framework—not as a standalone signal generator.